RICE UNIVERSITY

D1 · Collect: stream a Wikipedia dump into Parquet

Read article pages from compressed XML into a clean Parquet table, without loading the whole file.

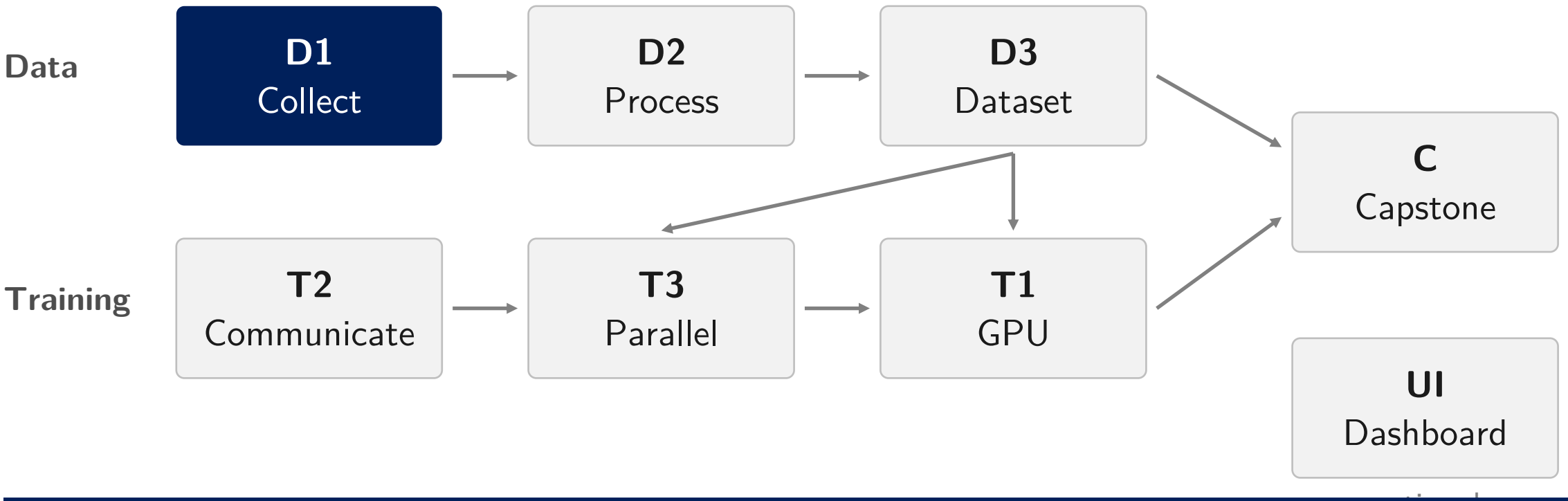
COMP 329 course staff

Rice University

Spring 2027 · Assignment D1

Release week 1 (Fri) · Checkpoint week 2 · Due week 3 · 7% of grade

Where this assignment fits



In our pipeline

This assignment: `data/tiny-wiki.xml.bz2` → **D1** → `articles.parquet`

What you will build, and why

Turn a raw XML dump into a typed table the next stage can trust, and explain every page you skip.

1 **Streaming**

Real dumps are far larger than memory. Process one page, release it, keep only a small batch.

2 **Provenance**

Every row keeps its page ID, revision, timestamp, source and license.

3 **Accountability**

Each skipped page gets one logged reason, so counts always add up.

What goes in, what comes out



Files you produce

`articles.parquet`
Accepted article records

`articles.rejected.jsonl`
One line per skipped page, with its reason

`articles.stats.json`
Counts, sizes, throughput, checksum

Streaming keeps memory bounded

1

Open stream

Pick open, gzip.open or bz2.open from the suffix; read bytes as a stream.

→

2

iterparse

Handle each </page> end event as soon as it completes.

→

3

Extract + check

Page fields, revision fields, then the first matching skip reason.

→

4

Batch

Hold at most `batch_size` accepted records.

→

5

Write + clear

Write a Parquet row group; clear the element and root references.

Key idea

Without clearing finished elements, incremental parsing can still keep the whole tree in memory.

Page identity vs. revision identity

```
<page>
  <title>Arrays</title>
  <ns>0</ns>
  <id>11</id>
  <revision>
    <id>101</id>
    <timestamp>2027-01-01…</timestamp>
    <text>Arrays store…</text>
  </revision>
</page>
```

1. `doc_id = "11"`

   Comes from the page's own <id>, not the revision's.

2. `revision_id = "101"`

   Comes from the <id> inside <revision>.

3. **Skip reasons, first match wins**

   namespace → redirect → empty → `missing_metadata`

4. `pages_seen` counts everything

   Skipped pages count too; `max_pages` limits this number.

What a correct first job produces

37

pages seen

34

articles kept

3

pages skipped

Key idea

The three skips: one redirect, one non-article namespace, one empty page. The short article and the contact-spam article stay; filtering them is D2's job.

What you write, what you get

You write

`collect(...)`

`starter/collect.py` · 7 numbered TODOs, keep the signature

Provided

- SCHEMA and `strip_markup` (mwparserfromhell)
- `file_hash` and `write_json`
- The lab.py launcher (spawn-safe imports)
- Public tests in `tests/test_d1.py`

Ground rules

Keep the signatures · don't edit `starter/pipeline_lab` · run through `course.py` or `lab.py`.

Build it in this order

1 — 2 — 3 — 4 — 5

1 One page

Parse one uncompressed page; test page vs. revision IDs.

2 Skip + log

Counters and the rejection log; clear finished pages.

3 Batches

Bounded Parquet row groups; flush leftovers at the end.

4 Compression

Stream .xml, .gz and .bz2 in binary mode.

5 Safe + parallel

Temp files, rename on success, spawned workers.

```
check d1 --checkpoint
```

```
check d1
```

Check

Done when: All 5 tests pass, and the first job reports 37 / 34 / 3.

If something goes wrong

`doc_id is 10001 for page 1`

Check: You used the revision ID instead of the page ID.

Final rows disappear

Check: Flush the last, partially filled batch after parsing.

Memory grows with file length

Check: Clear completed elements and root references.

Broken XML changes the output

Check: Rename temp files only after parsing succeeds.

Two-worker run hangs

Check: Use the launcher, a spawn context, and shut the pool down in finally.

Counts do not add up

Check: Every page goes to `pages_seen`; each skip has exactly one reason.

The performance study

What limits throughput, and how long would the full dump take?

Predict first

Decompression, XML traversal, writing and setup stay serial. Predict before timing.

Workers 1 / 2 / 4
37 and 96 pages

Decompression alone
find the bottleneck

Full English dump
time and cost estimate

How to measure: One warmup + three repeats. Scale your best compressed MB/s to the full dump; state the size and price you assume.

How you are graded

20	40	16	24
Checkpoint	Final correctness	Performance	Analysis

Correctness: schema/provenance 10 · filtering/counters 8 · streaming/compression 10 · rollback/parallel 12

Check

Done when: All 5 tests pass, and the first job reports 37 / 34 / 3.

Your first ten minutes

```
$ bash setup.sh
$ source .venv/bin/activate
$ python course.py preflight
$ python course.py inputs --output runs/inputs
$ python course.py check d1 --checkpoint
$ python lab.py d1 --input data/tiny-wiki.xml.bz2 \
    --output runs/d1-first --workers 1
```

Your folder

`01_d1_collect/`

Handout:
`01_d1_collect/README.md`